

Numerical Analysis for PDE's

Lecture 7

Finite Volume Method in 2D:

Straight and skewed boundaries,
Properties of 2D FVM Laplacian matrix

Finite Volume Method in 2D

Steady-state diffusion equation

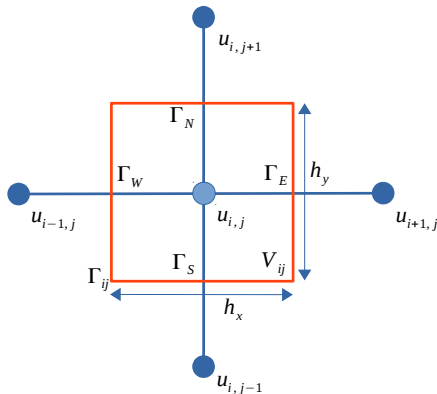
$$-\nabla \cdot (k \nabla u) = f, \quad (x, y) \in \Omega$$

$$-k \frac{\partial u}{\partial \mathbf{n}} = \alpha(u - u_0), \quad (x, y) \in \partial\Omega$$

where $\Omega \subset \mathbb{R}^2$ is a two-dimensional domain; $k(x, y)$ and $f(x, y)$ are given functions; α and u_0 are given constants; and $\mathbf{n}$ is the outwards normal to the boundary $\partial\Omega$.

Vertex-Centered Finite-Volume

The finite volume Ω_{ij} with the area V_{ij} is centered around the vertex (x_i, y_j) and has the boundary Γ_{ij}



Integrating over cell (common procedure)

$$\iint_{V_{ij}} [-\nabla \cdot (k \nabla u)] dV = \iint_{V_{ij}} f dV$$

Applying the 2D Divergence Theorem

$$\oint_{\Gamma_{ij}} (-k \nabla u \cdot \mathbf{n}) d\Gamma = \iint_{V_{ij}} f dV$$

$$\oint_{\Gamma_{ij}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma = \iint_{V_{ij}} f dV$$

Divide by the area (volume) of the cell V_{ij} for true averaging. Can also be done at the end.

Single-point approximation for the right-hand side

$$\iint_{V_{ij}} f \, dV \approx f_{ij} h_x h_y$$

- For rectangular cells the area (volume) is $h_x h_y$, for cells of other shapes use the corresponding formula.
- For some non-uniform grids, the vertex point (x_i, y_j) is not in the middle point of the integration cell V_{ij} , which lowers the approximation order (see 1D case, Lecture 6).

Splitting boundary in segments

$$\oint_{\Gamma_{ij}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma = \int_{\Gamma_W} \left(-k \left(-\frac{\partial u}{\partial x} \right) \right) dy + \int_{\Gamma_E} \left(-k \frac{\partial u}{\partial x} \right) dy \\ + \int_{\Gamma_S} \left(-k \left(-\frac{\partial u}{\partial y} \right) \right) dx + \int_{\Gamma_N} \left(-k \frac{\partial u}{\partial y} \right) dx$$

Pay attention to the direction of the unit normal vector $\mathbf{n}$ at each boundary. Easy with rectangular cells:

- North boundary Γ_N

$$\frac{\partial u}{\partial \mathbf{n}} = \nabla u \cdot \mathbf{n} = \left\langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right\rangle \cdot \langle 0, 1 \rangle = \frac{\partial u}{\partial y}$$

- West boundary Γ_W

$$\frac{\partial u}{\partial \mathbf{n}} = \nabla u \cdot \mathbf{n} = \left\langle \frac{\partial u}{\partial x}, \frac{\partial u}{\partial y} \right\rangle \cdot \langle -1, 0 \rangle = -\frac{\partial u}{\partial x}$$

Central-difference and single-point approximation for fluxes

Approximation of the integral brings you to an edge of the main grid. FD approximation of the partial derivative connects you with the vertex values of the unknown function.

$$\int_{\Gamma_W} k \frac{\partial u}{\partial x} dy \approx h_y k \frac{\partial u}{\partial x} \Big|_{(x_{i-1/2}, y_j)} \approx h_y k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x}$$

$$\int_{\Gamma_E} \left(-k \frac{\partial u}{\partial x} \right) dy \approx -h_y k \frac{\partial u}{\partial x} \Big|_{(x_{i+1/2}, y_j)} \approx -h_y k_{i+1/2,j} \frac{u_{i+1,j} - u_{i,j}}{h_x}$$

$$\int_{\Gamma_S} k \frac{\partial u}{\partial y} dx \approx h_x k \frac{\partial u}{\partial y} \Big|_{(x_i, y_{j-1/2})} \approx h_x k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y}$$

$$\int_{\Gamma_N} \left(-k \frac{\partial u}{\partial y} \right) dx \approx -h_x k \frac{\partial u}{\partial y} \Big|_{(x_i, y_{j+1/2})} \approx -h_x k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y}$$

Inner point FVM equation

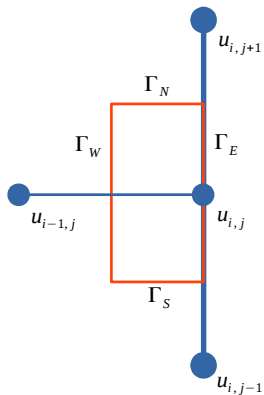
$$h_y k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} - h_y k_{i+1/2,j} \frac{u_{i+1,j} - u_{i,j}}{h_x} \\ + h_x k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y} - h_x k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y} = h_x h_y f_{ij}$$

Rearrange to recognize the system matrix entries and divide by $h_x h_y$ for proper averaging:

$$- \frac{k_{i-1/2,j}}{h_x^2} u_{i-1,j} - \frac{k_{i,j-1/2}}{h_y^2} u_{i,j-1} \\ + \left(\frac{k_{i-1/2,j}}{h_x^2} + \frac{k_{i,j-1/2}}{h_y^2} + \frac{k_{i+1/2,j}}{h_x^2} + \frac{k_{i,j+1/2}}{h_y^2} \right) u_{i,j} \\ - \frac{k_{i+1/2,j}}{h_x^2} u_{i+1,j} - \frac{k_{i,j+1/2}}{h_y^2} u_{i,j+1} = f_{ij}$$

Straight and skewed boundaries

Neumann and Robin BC's: straight boundaries



BC's at straight boundaries

Integrating the equation over the boundary-adjacent half-cell and applying the 2D Divergence Theorem:

$$\oint_{\Gamma_{ij}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma = \iint_{V_{ij}} f dV$$

Splitting the boundary in segments:

$$\begin{aligned} \oint_{\Gamma_{ij}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma &= \int_{\Gamma_W} \left(-k \left(-\frac{\partial u}{\partial x} \right) \right) dy + \int_{\Gamma_E} \left(-k \frac{\partial u}{\partial x} \right) dy \\ &\quad + \int_{\Gamma_S} \left(-k \left(-\frac{\partial u}{\partial y} \right) \right) dx + \int_{\Gamma_N} \left(-k \frac{\partial u}{\partial y} \right) dx \end{aligned}$$

Substituting the boundary condition $-k \frac{\partial u}{\partial \mathbf{n}} = \alpha(u - u_0)$ in the flux through the corresponding edge:

$$\int_{\Gamma_E} \left(-k \frac{\partial u}{\partial x} \right) dy = \int_{\Gamma_E} \alpha(u - u_0) dy$$

BC's at straight boundaries: approximations

$$\int_{\Gamma_W} k \frac{\partial u}{\partial x} dy \approx h_y k \frac{\partial u}{\partial x} \Big|_{(x_{i-1/2}, y_j)} \approx h_y k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x}$$

$$\int_{\Gamma_E} \alpha(u - u_0) dy \approx h_y \alpha_{i,j} (u_{i,j} - u_0)$$

$$\int_{\Gamma_S} k \frac{\partial u}{\partial y} dx \approx \frac{h_x}{2} k \frac{\partial u}{\partial y} \Big|_{(x_i, y_{j-1/2})} \approx \frac{h_x}{2} k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y}$$

$$\int_{\Gamma_N} \left(-k \frac{\partial u}{\partial y} \right) dx \approx -\frac{h_x}{2} k \frac{\partial u}{\partial y} \Big|_{(x_i, y_{j+1/2})} \approx -\frac{h_x}{2} k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y}$$

$$\iint_{V_{ij}} f dV \approx \frac{h_x h_y}{2} f_{i,j}$$

Boundary point FVM equation (straight boundary)

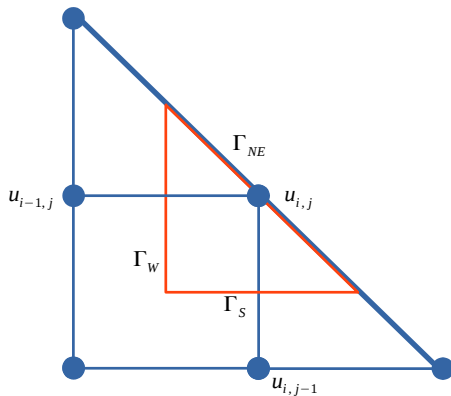
$$h_y k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} + h_y \alpha_{i,j} (u_{i,j} - u_0) \\ + \frac{h_x}{2} k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y} - \frac{h_x}{2} k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y} = \frac{h_x h_y}{2} f_{ij}$$

Rearranging and dividing by $h_x h_y / 2$:

$$- \frac{2k_{i-1/2,j}}{h_x^2} u_{i-1,j} - \frac{2k_{i,j-1/2}}{2h_y^2} u_{i,j-1} \\ + 2 \left(\frac{k_{i-1/2,j}}{h_x^2} + \frac{k_{i,j-1/2}}{2h_y^2} + \frac{\alpha_{i,j}}{h_x} + \frac{k_{i,j+1/2}}{2h_y^2} \right) u_{i,j} \\ - \frac{2k_{i,j+1/2}}{2h_y^2} u_{i,j+1} = f_{ij} + \frac{2\alpha_{i,j}}{h_x} u_0$$

where (x_i, y_j) is an Eastern boundary point.

Skewed boundaries: vertex at the boundary



BC's at skewed boundaries

Integrating the equation over the boundary-adjacent triangular cell and applying the 2D Divergence Theorem:

$$\oint_{\Gamma_{ij}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma = \iint_{V_{ij}} f dV$$

Splitting the boundary in three segments:

$$\begin{aligned} \oint_{\Gamma_{ij}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma &= \int_{\Gamma_W} \left(-k \left(-\frac{\partial u}{\partial x} \right) \right) dy + \int_{\Gamma_S} \left(-k \left(-\frac{\partial u}{\partial y} \right) \right) dx \\ &\quad + \int_{\Gamma_{NE}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma \end{aligned}$$

Substituting the boundary condition $-k \frac{\partial u}{\partial \mathbf{n}} = \alpha(u - u_0)$ in the flux at the skewed edge:

$$\int_{\Gamma_{NE}} \left(-k \frac{\partial u}{\partial \mathbf{n}} \right) d\Gamma = \int_{\Gamma_{NE}} \alpha(u - u_0) d\Gamma$$

BC's at skewed boundaries: approximations

$$\int_{\Gamma_W} k \frac{\partial u}{\partial x} dy \approx h_y k \frac{\partial u}{\partial x} \Big|_{(x_{i-1/2}, y_j)} \approx h_y k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x}$$

$$\int_{\Gamma_S} k \frac{\partial u}{\partial y} dx \approx h_x k \frac{\partial u}{\partial y} \Big|_{(x_i, y_{j-1/2})} \approx h_x k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y}$$

$$\int_{\Gamma_{NE}} \alpha(u - u_0) d\Gamma \approx \sqrt{h_x^2 + h_y^2} \alpha_{i,j} (u_{i,j} - u_0)$$

$$\iint_{V_{ij}} f dV \approx \frac{h_x h_y}{2} f_{i,j}$$

Boundary point FVM equation (skewed boundary)

$$h_y k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} + h_x k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y} + \sqrt{h_x^2 + h_y^2} \alpha_{i,j} (u_{i,j} - u_0) = \frac{h_x h_y}{2} f_{ij}$$

Rearranging and dividing by $h_x h_y / 2$:

$$\begin{aligned} & - \frac{2k_{i-1/2,j}}{h_x^2} u_{i-1,j} - \frac{2k_{i,j-1/2}}{h_y^2} u_{i,j-1} \\ & + 2 \left(\frac{k_{i-1/2,j}}{h_x^2} + \frac{k_{i,j-1/2}}{h_y^2} + \frac{\alpha_{i,j} \sqrt{h_x^2 + h_y^2}}{h_x h_y} \right) u_{i,j} \\ & = f_{ij} + \frac{2\alpha_{i,j} \sqrt{h_x^2 + h_y^2}}{h_x h_y} u_0 \end{aligned}$$

where (x_i, y_j) is a North-Eastern skewed-boundary point.

Properties of the 2D FVM Laplacian matrix

Symmetry of the FVM matrix with Dirichlet BC's

- 1 The matrix A is symmetric if $A^T = A$, i.e. $a_{nm} = a_{mn}$
- 2 Alternatively, A is symmetric if $\mathbf{v}^T A \mathbf{u} = (\mathbf{v}^T A \mathbf{u})^T = \mathbf{u}^T A^T \mathbf{v} = \mathbf{u}^T A \mathbf{v}$, for all $\mathbf{u}, \mathbf{v} \in \mathbb{R}^{N-1}$ satisfying homogeneous Dirichlet BC's
- 3 The FVM equations have the form:

$$[A\mathbf{u}]_{i,j} = k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} - k_{i+1/2,j} \frac{u_{i+1,j} - u_{i,j}}{h_x^2} \\ + k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y^2} - k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y^2}$$

- 4 Homogeneous Dirichlet BC's (rectangular domain):

$$u_{0,j} = u_{N_x,j} = 0, \quad j = 1, \dots, N_y - 1$$

$$u_{i,0} = u_{i,N_y} = 0, \quad i = 1, \dots, N_x - 1$$

Proving symmetry

For any fixed $j = 1, \dots, N_y - 1$:

$$\begin{aligned} & \sum_{i=1}^{N_x-1} v_{i,j} \left[k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} - k_{i+1/2,j} \frac{u_{i+1,j} - u_{i,j}}{h_x^2} \right] \\ &= \sum_{i=1}^{N_x-1} v_{i,j} k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} - \sum_{i=1}^{N_x-1} v_{i,j} k_{i+1/2,j} \frac{u_{i+1,j} - u_{i,j}}{h_x^2} \\ &= \sum_{i=1}^{N_x-1} v_{i,j} k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} - \sum_{i=2}^{N_x} v_{i-1,j} k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} \\ &= \sum_{i=1}^{N_x} v_{i,j} k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} - \sum_{i=1}^{N_x} v_{i-1,j} k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} \\ &= \sum_{i=1}^{N_x} \frac{v_{i,j} - v_{i-1,j}}{h_x} k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} \end{aligned}$$

Proving symmetry

Similarly, for any fixed $i = 1, \dots, N_x - 1$:

$$\begin{aligned} & \sum_{j=1}^{N_y-1} v_{i,j} \left[k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y^2} - k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y^2} \right] \\ &= \sum_{i=1}^{N_y} \frac{v_{i,j} - v_{i,j-1}}{h_y} k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y} \end{aligned}$$

Proving symmetry

$$\begin{aligned}\mathbf{v}^T \mathbf{A} \mathbf{u} &= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x-1} v_{i,j} \left[k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x^2} - k_{i+1/2,j} \frac{u_{i+1,j} - u_{i,j}}{h_x^2} \right. \\ &\quad \left. + k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y^2} - k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y^2} \right] \\ &= \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x} \frac{v_{i,j} - v_{i-1,j}}{h_x} k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} \\ &\quad + \sum_{j=1}^{N_y} \sum_{i=1}^{N_x-1} \frac{v_{i,j} - v_{i,j-1}}{h_y} k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y} = \mathbf{u}^T \mathbf{A} \mathbf{v}\end{aligned}$$

Last equality follows from the symmetry of the expression with respect to $\mathbf{u}$ and $\mathbf{v}$.

Positivity of the FVM matrix

- 1 Matrix A is positive definite if $\mathbf{u}^T A \mathbf{u} > 0$, for all $\mathbf{u} \neq \mathbf{0}$
- 2 For $k(x, y) > 0$, matrix produced by the FVM discretization is positive definite:

$$\begin{aligned} \mathbf{u}^T A \mathbf{u} = & \sum_{j=1}^{N_y-1} \sum_{i=1}^{N_x} k_{i-1/2,j} \left(\frac{u_{i,j} - u_{i-1,j}}{h_x} \right)^2 \\ & + \sum_{j=1}^{N_y} \sum_{i=1}^{N_x-1} k_{i,j-1/2} \left(\frac{u_{i,j} - u_{i,j-1}}{h_y} \right)^2 > 0 \end{aligned}$$

See 2D FDM proof for additional step. For other types of BC's, boundary-point equations need to be considered separately.

Diagonal dominance of the FVM matrix

- 1 Matrix A is diagonally dominant if $|a_{nn}| \geq \sum_{m \neq n} |a_{nm}|$
- 2 Matrix A is called a Z -matrix, if $a_{nm} \leq 0$, $m \neq n$
- 3 Matrix produced by the FVM discretization is a diagonally dominant Z -matrix
- 4 Some rows are strictly diagonally dominant, $|a_{nn}| > \sum_{m \neq n} |a_{nm}|$

Proving diagonal dominance

$$\begin{aligned} & h_y k_{i-1/2,j} \frac{u_{i,j} - u_{i-1,j}}{h_x} - h_y k_{i+1/2,j} \frac{u_{i+1,j} - u_{i,j}}{h_x} \\ & + h_x k_{i,j-1/2} \frac{u_{i,j} - u_{i,j-1}}{h_y} - h_x k_{i,j+1/2} \frac{u_{i,j+1} - u_{i,j}}{h_y} \\ & = -\frac{h_y}{h_x} k_{i-1/2,j} u_{i-1,j} - \frac{h_y}{h_x} k_{i+1/2,j} u_{i+1,j} - \frac{h_x}{h_y} k_{i,j-1/2} u_{i,j-1} - \frac{h_x}{h_y} k_{i,j+1/2} u_{i,j+1} \\ & + \left(\frac{h_y}{h_x} k_{i-1/2,j} + \frac{h_y}{h_x} k_{i+1/2,j} + \frac{h_x}{h_y} k_{i,j-1/2} + \frac{h_x}{h_y} k_{i,j+1/2} \right) u_{i,j} \end{aligned}$$

Hence, for $k(x, y) > 0$,

$$a_{nn} < 0, \quad n \neq m, \quad \text{and} \quad |a_{nn}| \geq \sum_{n \neq m} |a_{nm}|,$$

At boundary points: $|a_{nn}| > \sum_{n \neq m} |a_{nm}|$.

Positivity of the solution

- 1 If A is a weakly chained FVM matrix, then the solution of the discrete problem $A\mathbf{u} = \mathbf{f}$ with $\mathbf{f} \geq \mathbf{0}$ is positive, i.e. $\mathbf{u} \geq \mathbf{0}$
- 2 First show that A is invertible, i.e. $A\mathbf{u} = \mathbf{0}$, only if $\mathbf{u} = \mathbf{0}$. This can be proven using the facts that: (1) A is a diagonally dominant Z matrix with at least some rows that are strictly diagonally dominant (boundary points), and (2) A is a weakly chained diagonally dominant matrix, i.e., there is a directed walk from each inner grid point to any boundary point via the inner grid edges.
- 3 Since A is an invertible positive Z matrix, A is an M matrix.
- 4 Since A is an M matrix, its inverse A^{-1} is a positive matrix, $A^{-1} \geq 0$, i.e., all entries of A^{-1} are positive.
- 5 Since A^{-1} is positive, A is a *monotone* matrix, i.e., $A\mathbf{u} \geq \mathbf{0}$ implies $\mathbf{u} \geq \mathbf{0}$

A Tool to Estimate the Location of Eigenvalues

Lemma (Gerschgorin Circles)

Let $A \in \mathbb{C}^{n \times n}$ be an arbitrary matrix with elements $a_{k,m}$. The spectrum $\sigma(A)$ is contained within the union of n disks:

$$\sigma(A) \subset \bigcup_{k=1}^n \mathbb{D}_k \quad (1)$$

where each Gerschgorin disk $\mathbb{D}_k$ is defined in the complex plane as:

$$\mathbb{D}_k = \left\{ z \in \mathbb{C} : |z - a_{k,k}| \leq \sum_{m \neq k}^n |a_{k,m}| \right\} \quad (2)$$

Boundary Property for Irreducible Matrices

If A is **irreducible** (e.g., connected grid graphs in FDM/FVM) and an eigenvalue $\lambda \in \sigma(A)$ lies on the boundary of the total union $\partial(\bigcup \mathbb{D}_k)$, then λ must lie on the boundary $\partial \mathbb{D}_k$ for **all** $k = 1, \dots, n$.

Example: FDM/FVM Laplacian, uniform mesh, $k = 1$

Inner point FVM equation:

$$-\frac{1}{h_x^2}u_{i-1,j} - \frac{1}{h_y^2}u_{i,j-1} + \left(\frac{2}{h_x^2} + \frac{2}{h_y^2}\right)u_{i,j} - \frac{1}{h_x^2}u_{i+1,j} - \frac{1}{h_y^2}u_{i,j+1} = f_{ij}$$

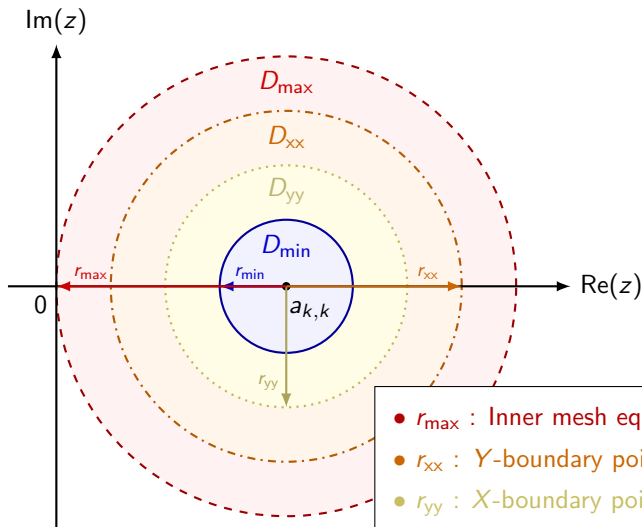
Since the main diagonal is constant, the centers of all Gerschgorin disks are the same:

$$a_{k,k} = \frac{2}{h_x^2} + \frac{2}{h_y^2}, \quad k = 1, \dots, (N_x - 1)(N_y - 1)$$

The radii of disks have four possible values:

$$r_{\min} = \frac{1}{h_x^2} + \frac{1}{h_y^2}, \quad r_{\max} = \frac{2}{h_x^2} + \frac{2}{h_y^2}$$
$$r_{yy} = \frac{1}{h_x^2} + \frac{2}{h_y^2}, \quad r_{xx} = \frac{2}{h_x^2} + \frac{1}{h_y^2}$$

Example: $h_x < h_y$



- $r_{\max}$: Inner mesh equations (touches 0)
- r_{xx} : Y-boundary points
- r_{yy} : X-boundary points
- $r_{\min}$: Corner boundary points

Bounds on minimal and maximal eigenvalues

- Since the matrix is symmetric, all eigenvalues are real – on the $\text{Re}(z)$ axis.
- The smallest eigenvalue is at the leftmost possible location along the $\text{Re}(z)$ axis. However, it cannot be at zero, since in that case all eigenvalues must be zero as we have an irreducible matrix! Gerschgorin's Lemma shows that $\lambda_{\min} > 0$
- The largest eigenvalue is at the rightmost location along $\text{Re}(z)$. The algebraic bound:

$$\lambda_{\max} \leq a_{k,k} + r_{\max} = \frac{4}{h_x^2} + \frac{4}{h_y^2}$$

agrees very well with the analytical result (doubly uniform mesh):

$$\lambda_{k_x, k_y} = \frac{4}{h^2} \left(\sin^2 \left(\frac{k_x \pi}{2N_x} \right) + \sin^2 \left(\frac{k_y \pi}{2N_y} \right) \right) \leq \frac{8}{h^2}$$